

# Immigration and Credit in America

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- Immigrants start life in the U.S. *without* a U.S. credit report
  - Any foreign credit history generally unobserved to U.S. lenders

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## What We Do

- Describe **selection** of Immigrants compared to Non-Immigrants in their creditworthiness
- Describe **assimilation** in creditworthiness & access to credit over life cycle
- How **dynamics** of life cycle outcomes vary by one year difference in age of immigration, (with fixed effects for birth year & ZIP5 geography)

# How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

## Data

- U.S. consumer credit reporting data matched with immigration data

## What We Do

- Describe **selection** of Immigrants & **assimilation** dynamics over life cycle
  - Immigrants vs. same-age Non-Immigrants & Immigrants arriving at different ages

## Immigrants (vs. Non-Immigrants), Immigrants Arriving At Older (vs. Younger) Ages:

- **Positive selection:** More creditworthy (once credit scored)
- Despite better creditworthiness, **lower long-term access** to auto loans & mortgages

## Timing of immigration best explains our results (“history dependence”)

# Our Paper Contributes To Labor Economics & Household Finance Literatures

- Prior work shows the large positive contribution of legal immigrants to U.S. economy (e.g., Azoulay et al., 2022; Bernstein et al., 2025)
- We show the **selection** of immigrants based on their credit behaviors, complementing work focusing on labor market outcomes (e.g., Borjas, 1985; Abramitzsky & Boustan, 2017)
- We show **assimilation** into the U.S. credit markets over life cycle, complementing work on labor & cultural assimilation (e.g., Borjas, 1985; Abramitzsky et al., 2014, 2020, 2021; Lubotsky, 2007; Bleakley & Chin, 2010; Doran, Geber, & Isen, 2022; Bailey et al., 2022)
- We contribute to the household finance literature that has largely not studied immigrants (e.g., Dobbie et al., 2021; Zillessen, 2022; Bakker et al., 2025; Gorback & Schubert, 2025)

## Roadmap For Presentation

1. Data & Institutional Details
2. Credit Market Entry
3. Creditworthiness
4. Credit Access
5. Mechanisms

## **Data & Institutional Details**

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## University of Chicago Booth TransUnion Consumer Credit Panel (BTCCP)

- Anonymized data sample of 1 in 10 consumers with U.S. credit reports
- Monthly data from 2000 to 2024
- Individual tradeline data (e.g., details on each account, including opening dates) & consumer-level data (e.g., birth date, credit score, ZIP code)

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(2025, *Journal of Economic Literature*)

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## **American Community Survey (ACS), 2010**

- Validate our immigration classification

## **Survey of Consumer Finances (SCF), 2022**

- Evaluate mechanisms behind our results

## Classifying Immigrants In The BTCCP

Before 2012, U.S. Social Security Administration assigned first 5 digits of the 9-digit Social Security Numbers (SSNs) sequentially within states (Puckett, 2009)

Consumers assigned SSNs as adults are very likely to be (legal) immigrants

- validated in prior work (Klopfer & Miller, 2024; Yonker, 2017; Doran, Gelber, & Isen, 2022; Bernstein et al., 2025)
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1. Create lookup table mapping first 5 digits of SSN to SSN assignment year (*SSN Year*)
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**Entrant Sample Of Birth Years 1975 to 1987 With SSN Ages 0 to 29:**

**6,122,932 consumers – 344,261 (5.6%) “Immigrants” (ACS: 6.6% includes illegal)**

Follow **credit outcomes** in BTCCP for **ages 18 to 40**

- Immigrant credit access may be adversely affected by **information frictions**:
  1. **Not having a credit report on arrival**
    - **Lender's prior:** Consumer with no U.S. credit report is **high credit risk**
  2. **After entering the credit system, have a shorter history than Non-Immigrants**
    - Length of credit report has positive impact to credit scores ( $\approx 15\%$  FICO)

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    - Length of credit report has positive impact to credit scores ( $\approx 15\%$  FICO)
- Different cultural **tastes** & informational barriers for consumers
  - Differences in familiarity & willingness to use debt, understanding of how to build credit score
- Lenders use credit scores (+ other information) in decisions
  - Can use immigration status, can't use protected characteristics (e.g., national origin, race) under Equal Credit Opportunity Act (ECOA)

## Credit Market Entry

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# Main Cross-Sectional Regression Specifications

$\beta_1$ : average difference for Immigrants vs. Non-Immigrants from OLS regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i \quad (1)$$

- Includes fixed effects: birth year ( $\mu_{b(i)}$ ) and geography based on first ZIP5 ( $\eta_{g(i)}$ )

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$\beta_2$ : marginal difference for SSN Age one year later, from OLS regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i \quad (2)$$

- Cluster standard errors by birth year

# Immigrants, & Older SSN Age Immigrants, Enter Credit Market Later Than Non-Immigrants & Younger SSN Age Immigrants

|                   | Age at First Credit Report |                   |
|-------------------|----------------------------|-------------------|
|                   | (1)                        | (2)               |
| Immigrant         | 4.98***<br>(0.14)          | 1.81***<br>(0.09) |
| SSN Age           |                            | 0.74***<br>(0.01) |
| F.E. Birth Year   | X                          | X                 |
| F.E. First Zip5   | X                          | X                 |
| $R^2$             | 0.264                      | 0.289             |
| Mean, SSN Age <21 | 19.86                      | 19.86             |

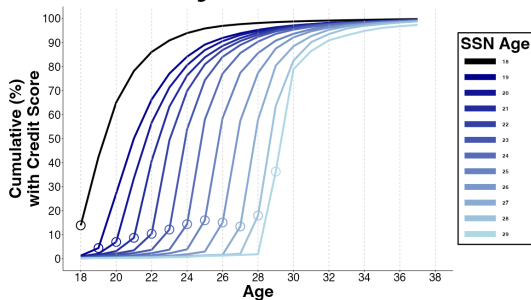
$N = 6,122,932$ , \*\*\* 0.5%, \*\*, 1%, \* 5% statistical significance

# Creditworthiness

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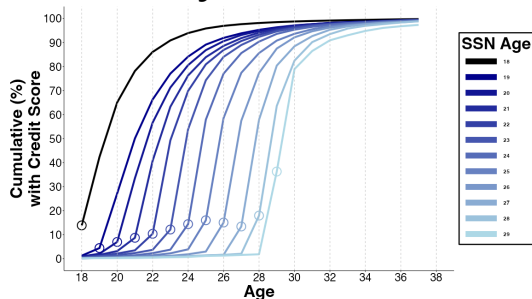
# Once Scored, Immigrants Have Higher Credit Scores Than Non-Immigrants

## A. Any Credit Score

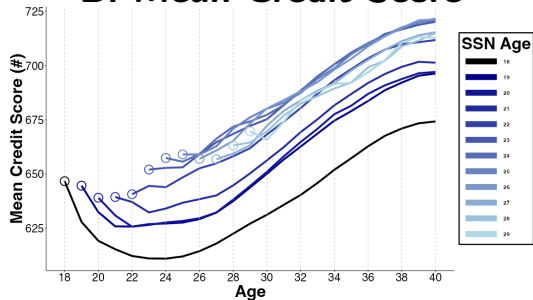


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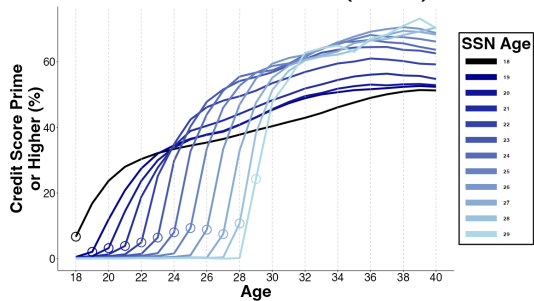


## B. Mean Credit Score



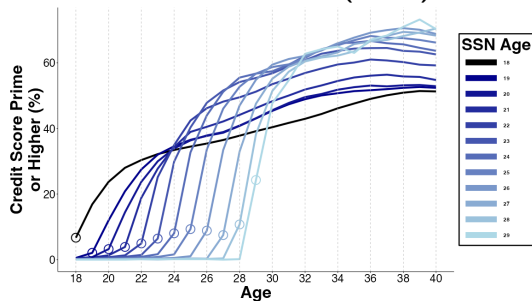
Immigrants (vs. Non-Immigrants) +27 points average scores by age 30, +33 by age 40

## C. Prime or Higher Credit Score (660+)



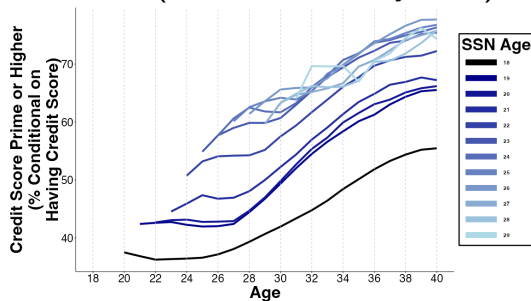
# Positive Selection Of Immigrants In Credit Score Distribution

## C. Prime or Higher Credit Score (660+)



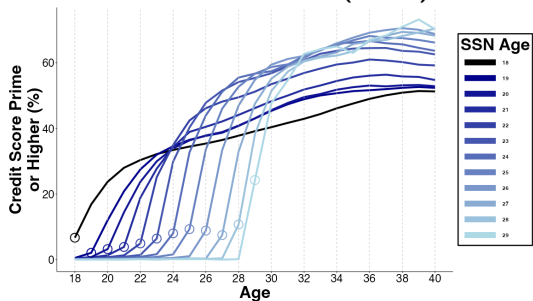
Immigrants:  $\downarrow$  Pr(subprime),  $\uparrow$  Pr(superprime)

## D. Prime or Higher Credit Score (Conditional On Any Score)

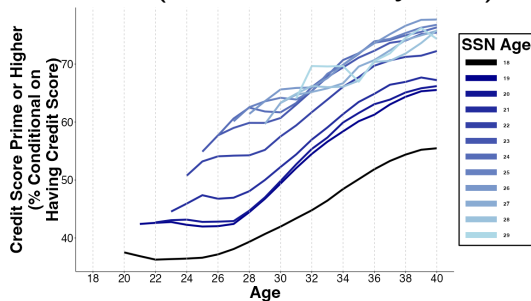


# Positive Selection Of Immigrants In Credit Score Distribution

## C. Prime or Higher Credit Score (660+)



## D. Prime or Higher Credit Score (Conditional On Any Score)



Immigrants:  $\downarrow$   $\Pr(\text{subprime})$ ,  $\uparrow$   $\Pr(\text{superprime})$

More creditworthy behaviors:  $\downarrow$  delinquency rates &  $\downarrow$  credit card utilization

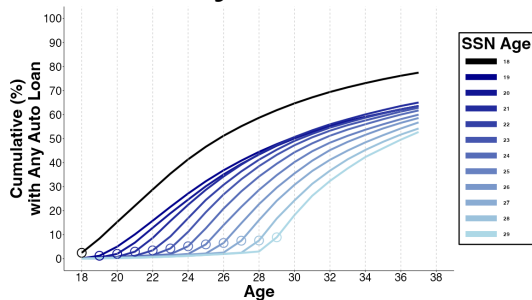
$\downarrow$  5% delinquency rates conditional on credit score

## Credit Access

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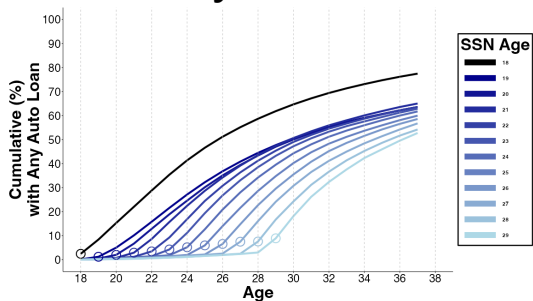
# Immigrants Persistently Less Likely To Hold Auto Loans (& Mortgages)

## A. Any Auto Loan

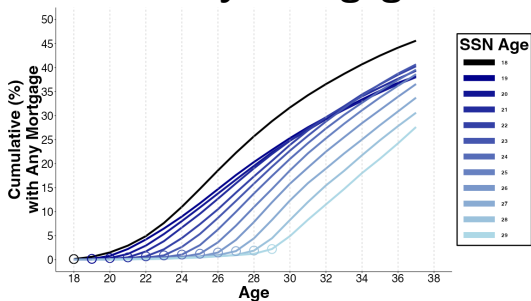


# Immigrants Persistently Less Likely To Hold Auto Loans (& Mortgages)

## A. Any Auto Loan



## B. Any Mortgage



# By Age 37, Immigrants 17% (16%) Less Likely To Have Auto Loan (Mortgage)

- ↓ Auto Loan & Mortgage Use For Later In Life Immigrants

| Dep. Var: By Age 37, has ... | <u>Auto Loan</u>    |                    | <u>Mortgage</u>    |                    | <u>Credit Card</u> |                |
|------------------------------|---------------------|--------------------|--------------------|--------------------|--------------------|----------------|
|                              | (1)                 | (2)                | (3)                | (4)                | (5)                | (6)            |
| Immigrant                    | -12.99***<br>(0.80) | -6.87***<br>(0.46) | -7.18***<br>(0.74) | 0.39<br>(0.53)     | 0.31<br>(0.22)     | 0.15<br>(0.16) |
| SSN Age                      |                     | -1.44***<br>(0.10) |                    | -1.77***<br>(0.09) |                    | 0.04<br>(0.03) |
| F.E. Birth Year              | X                   | X                  | X                  | X                  | X                  | X              |
| F.E. First Zip5              | X                   | X                  | X                  | X                  | X                  | X              |
| $R^2$                        | 0.039               | 0.039              | 0.062              | 0.063              | 0.025              | 0.025          |
| Mean, SSN Age <21            | 77.22               | 77.22              | 45.46              | 45.46              | 94.93              | 94.93          |

$N = 6,122,932$ , \*\*\* 0.5%, \*\*, 1%, \* 5% statistical significance

## “Paired Cohorts”: Comparing Immigrants vs. One Year Younger SSN Age (Immigrants With Same Birth Year)

$$Y_{k(p),t} = \sum_{\tau=-1}^{+13} \delta_{\tau} (\mathbb{I}\{\text{focal cohort}\}_{k(p)} \times \mathbb{I}\{t = \tau\}) + \pi_{k(p)} + \nu_{p,t} + \epsilon_{k(p),t} \quad (3)$$

- ‘Focal Cohort’:  $SSN\ Age_c$ , 22-30
- ‘Control Cohort’:  $SSN\ Age_c - 1$ , same birth year as focal. 68 pairs ( $p$ )
- 2,176 cohort-by-year observations ( $k(p), t$ ), s.e. clustered by birth year

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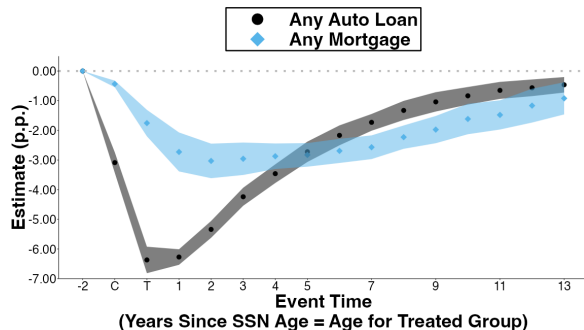
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- 2,176 cohort-by-year observations ( $k(p), t$ ), s.e. clustered by birth year
- **Results:**

**Any Auto Loan after 10 years:**

−0.46 p.p., −0.42 cumulative years

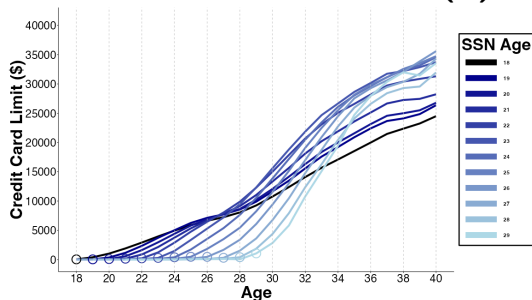
**Any Mortgage after 10 years:**

−0.92 p.p., −0.29 cumulative years



# Immigrants' Credit Card Limits Are Initially Lower Than Non-Immigrants, But Become Higher Than Non-Immigrants Over Life Cycle

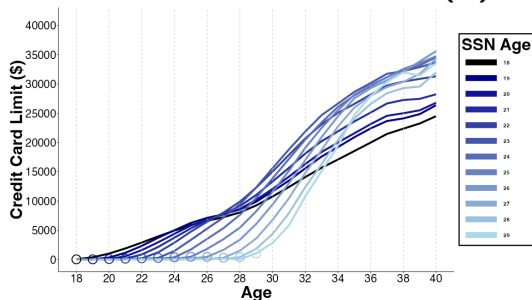
## A. Credit Card Limits (\$)



Similar patterns for number of credit cards,  
credit card limit per card

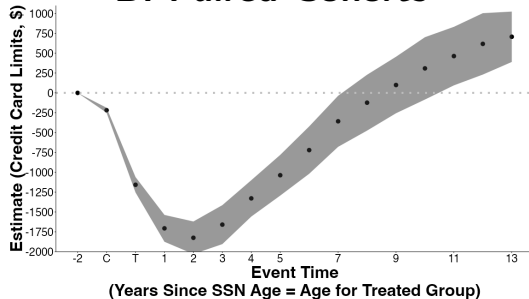
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## B. Paired Cohorts



**SSN Age +2 Years:**

−\$1,825 (s.e. \$105)

**SSN Age +13 Years:**

+\$707 (s.e. \$162)

# Mechanisms

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# What Mechanisms Explain?

- Less Credit For More Creditworthy Immigrants
- Less Credit For Later In Life Immigrants

**Creditworthiness**

**Emigration**

**Credit Demand & Tastes**

**History Dependence**

- Credit History Dependence: Financial friction of no U.S. credit report & shorter history
- Non-Credit History Dependence: less time in U.S. means...
  - fewer years of U.S. earnings
  - less far along in American life cycle

# Immigrants Are Averse To Purchasing Cars With Debt (SCF, 2022)

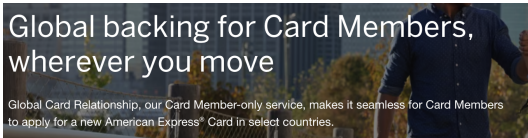
## -19% Immigrants Vs. Non-Immigrants

| Dep Var:                  | <u>Auto Debt Attitude</u> |                     | <u>Any Auto Debt</u> |                  | <u>Any Auto Debt</u> |                    |
|---------------------------|---------------------------|---------------------|----------------------|------------------|----------------------|--------------------|
|                           | (1)                       | (2)                 | (3)                  | (4)              | (5)                  | (6)                |
| Immigrant                 | -15.41***<br>(4.58)       | -23.02***<br>(7.25) | -4.27<br>(5.08)      | -11.14<br>(8.27) | -0.21<br>(5.01)      | -5.10<br>(7.87)    |
| Immigration Age           |                           | 0.98<br>(0.62)      |                      | 0.88<br>(0.76)   |                      | 0.63<br>(0.69)     |
| Auto Debt Attitude        |                           |                     |                      |                  | 26.34***<br>(3.23)   | 26.24***<br>(3.24) |
| Demographic Controls      | X                         | X                   | X                    | X                | X                    | X                  |
| $R^2$                     | 0.049                     | 0.050               | 0.063                | 0.063            | 0.106                | 0.106              |
| Mean, Immigration Age <21 | 79.58                     | 79.58               | 47.14                | 47.14            | 47.14                | 47.14              |

$N = 1,813$ , \*\*\* 1%, \*\* 5%, \* 10% statistical significance

# Non-Portability Of Credit Histories Across National Boundaries May Prevent Profitable Lending. Recent Market Solutions To Increase Credit Access:

## American Express Global Card Relationship



## Nova Credit

### International credit data

Credit Passport® gives you access to foreign consumer-permissioned credit data that's translated to a local equivalent score to serve the 2 million immigrants that arrive in the United States each year and 280 million immigrants worldwide.

## Equifax Canada Global Consumer Credit File

### Global Consumer Credit File

Helping you grow and retain your clients who are newcomers to Canada

Global Consumer Credit File provides lenders with a seamless and secure platform that connects with global credit bureaus and enables accurate and informed decisioning through robust data.

# Conclusions

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# How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

Study **selection & assimilation dynamics** over life cycle of Immigrants vs. Non-Immigrants  
- exploit variation in age of immigration in credit data matched with immigration information

**Immigrants vs. same age Non-Immigrants & Immigrants arriving at older ages:**

- **Positively selected:** more creditworthy (once scored)
- Despite better creditworthiness, **lower long-term access** to auto loans & mortgages
  - Lower mortgage access explained by timing of immigration
  - Lower auto loan access also due to auto loan-specific debt aversion
- *Timing* of immigration best explains our results (“**history dependence**”)

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Thank You!

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# Entrant Sample In BTCCP

## Sample Restrictions

- Birth years 1975 - 1987
- SSN Ages 0 to 29 (21 to 29 “Immigrants”)
- Credit report by 2011

## Entrant Sample

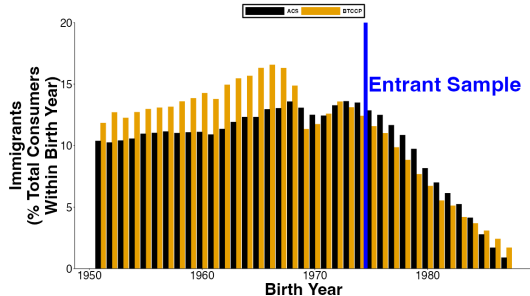
**6,122,932 consumers**

**– 344,261 (5.6%) “Immigrants” (ACS: 6.6% includes illegal)**

**Follow credit outcomes in BTCCP for ages 18 to 40**

## Distribution of Immigrants by...

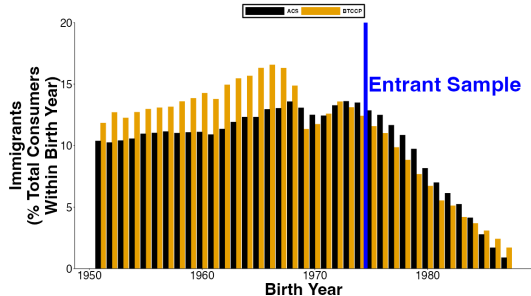
### A. Birth Year



# Immigrant Classification Aligns With American Community Survey (ACS) 🚀

## Distribution of Immigrants by...

### A. Birth Year



### B. Immigration Age



BTCCP: 11.27% SSN Age 21+

ACS: 10.2% Immigrated at Age 21+

Distribution of immigrants aligns by geography. Immigrants more mobile than non-immigrants.

## **“Entrant Sample”: Restrict To Birth Years 1975 - 1987 To Observe U.S. Entry At Ages 21 To 29 & Credit Outcomes For Ages 18 To 40 🍷**

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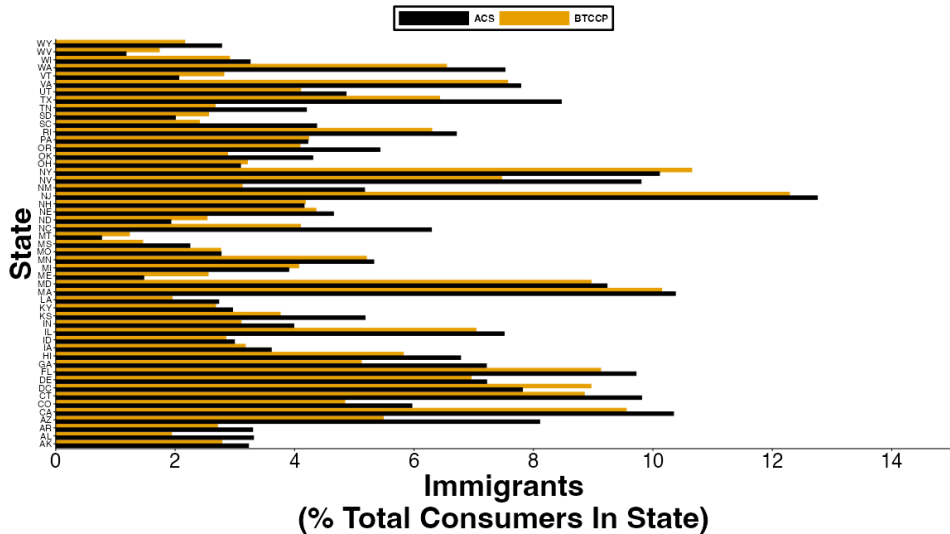
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|                                               |                    |
|-----------------------------------------------|--------------------|
| TransUnion Consumers                          |                    |
| ... born 1951 to 2004                         | 31,869,445         |
| ... with a SSN in TransUnion                  | 25,237,917         |
| <br>Clean Sample                              | <br>18,572,654     |
| ... SSN Age <21                               | 16,478,940         |
| ... SSN Age 21+                               | 2,093,714 (11.27%) |
| ... Immigrant Cohorts (Birth Year x SSN Year) | 775                |
| <br>Entrant Sample (SSN Age < 30)             | <br>6,122,932      |
| ... SSN Age < 21                              | 5,778,671          |
| ... SSN Age 21-29                             | 344,261 (5.62%)    |
| ... Immigrant Cohorts (Birth Year x SSN Year) | 102                |

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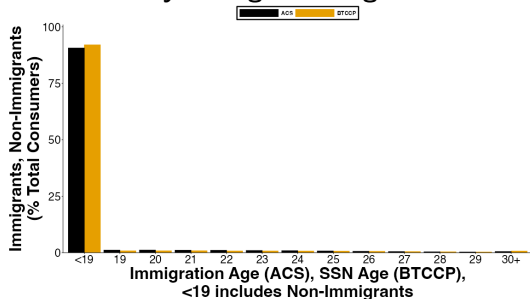
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# Geographic Comparison of Immigrants: ACS vs. BTCCP 🚀

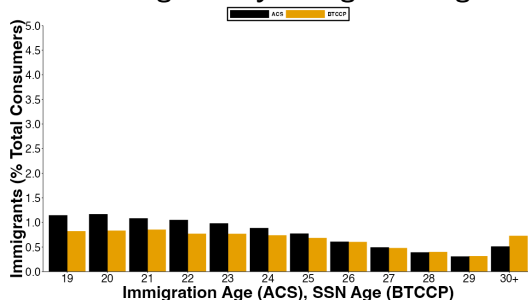


# Immigrant Classification in Entrants Sample: ACS vs. BTCCP

## A. Non-Immigrants and Immigrants by Immigration Age



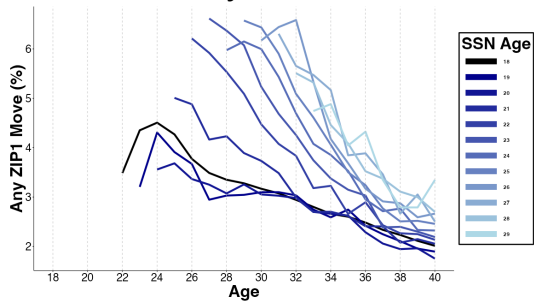
## B. Immigrants by Immigration Age



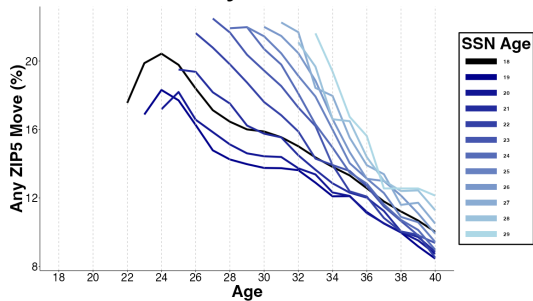
**BTCCP: 5.62% SSN Age 21 to 29**

**ACS: 6.57% Immigrated at Age 21 to 29**

## A. Any ZIP1 Move

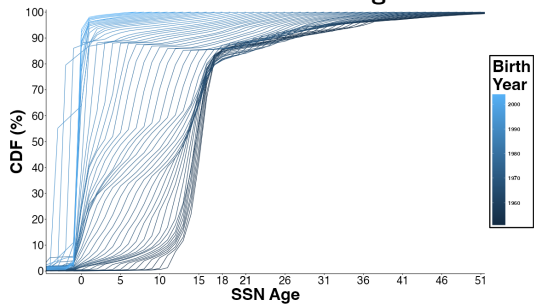


## B. Any ZIP5 Move

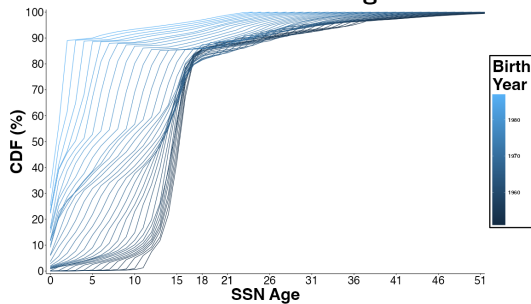


# SSN Ages By Birth Year

**A. Before Cleaning**



**B. After Cleaning**



# Consumers By SSN Age

| SSN Age | Clean Sample | Entrant Sample |
|---------|--------------|----------------|
| <19     | 16,155,157   | 5,677,001      |
| 19      | 175,194      | 50,475         |
| 20      | 148,589      | 51,195         |
| 21      | 148,019      | 52,450         |
| 22      | 141,492      | 47,302         |
| 23      | 145,218      | 47,214         |
| 24      | 149,000      | 45,314         |
| 25      | 147,899      | 42,005         |
| 26      | 142,319      | 36,976         |
| 27      | 133,147      | 29,383         |
| 28      | 123,338      | 24,415         |
| 29      | 113,435      | 19,202         |
| 30+     | 849,847      | 44,642         |

# Sample Frame and Timing of SSN Assignment

| Calendar<br>year | Birth year |      |      |      |      |      |      |      |      |      |      |      |      |
|------------------|------------|------|------|------|------|------|------|------|------|------|------|------|------|
|                  | 1975       | 1976 | 1977 | 1978 | 1979 | 1980 | 1981 | 1982 | 1983 | 1984 | 1985 | 1986 | 1987 |
| 1993             | ≤18        |      |      |      |      |      |      |      |      |      |      |      |      |
| 1994             | 19         | ≤18  |      |      |      |      |      |      |      |      |      |      |      |
| 1995             | 20         | 19   |      |      |      |      |      |      |      |      |      |      |      |
| 1996             | 21         | 20   | ≤18  |      |      |      |      |      |      |      |      |      |      |
| 1997             | 22         | 21   | 20   | 19   | ≤18  |      |      |      |      |      |      |      |      |
| 1998             | 23         | 22   | 21   | 20   | 19   | ≤18  |      |      |      |      |      |      |      |
| 1999             | 24         | 23   | 22   | 21   | 20   | 19   | ≤18  |      |      |      |      |      |      |
| 2000             | 25         | 24   | 23   | 22   | 21   | 20   | 19   | ≤18  |      |      |      |      |      |
| 2001             | 26         | 25   | 24   | 23   | 22   | 21   | 20   | 19   | ≤18  |      |      |      |      |
| 2002             | 27         | 26   | 25   | 24   | 23   | 22   | 21   | 20   | 19   | ≤18  |      |      |      |
| 2003             | 28         | 27   | 26   | 25   | 24   | 23   | 22   | 21   | 20   | 19   | ≤18  |      |      |
| 2004             | 29         | 28   | 27   | 26   | 25   | 24   | 23   | 22   | 21   | 20   | 19   | ≤18  |      |
| 2005             | 30         | 29   | 28   | 27   | 26   | 25   | 24   | 23   | 22   | 21   | 20   | 19   | ≤18  |
| 2006             | 31         | 30   | 29   | 28   | 27   | 26   | 25   | 24   | 23   | 22   | 21   | 20   | 19   |
| 2007             | 32         | 31   | 30   | 29   | 28   | 27   | 26   | 25   | 24   | 23   | 22   | 21   | 20   |
| 2008             | 33         | 32   | 31   | 30   | 29   | 28   | 27   | 26   | 25   | 24   | 23   | 22   | 21   |
| 2009             | 34         | 33   | 32   | 31   | 30   | 29   | 28   | 27   | 26   | 25   | 24   | 23   | 22   |
| 2010             | 35         | 34   | 33   | 32   | 31   | 30   | 29   | 28   | 27   | 26   | 25   | 24   | 23   |
| 2011             | 36         | 35   | 34   | 33   | 32   | 31   | 30   | 29   | 28   | 27   | 26   | 25   | 24   |
| 2012             | 37         | 36   | 35   | 34   | 33   | 32   | 31   | 30   | 29   | 28   | 27   | 26   | 25   |
| 2013             | 38         | 37   | 36   | 35   | 34   | 33   | 32   | 31   | 30   | 29   | 28   | 27   | 26   |
| 2014             | 39         | 38   | 37   | 36   | 35   | 34   | 33   | 32   | 31   | 30   | 29   | 28   | 27   |
| 2015             | 40         | 39   | 38   | 37   | 36   | 35   | 34   | 33   | 32   | 31   | 30   | 29   | 28   |
| 2016             | 41         | 40   | 39   | 38   | 37   | 36   | 35   | 34   | 33   | 32   | 31   | 30   | 29   |
| 2017             | 42         | 41   | 40   | 39   | 38   | 37   | 36   | 35   | 34   | 33   | 32   | 31   | 30   |
| 2018             | 43         | 42   | 41   | 40   | 39   | 38   | 37   | 36   | 35   | 34   | 33   | 32   | 31   |
| 2019             | 44         | 43   | 42   | 41   | 40   | 39   | 38   | 37   | 36   | 35   | 34   | 33   | 32   |
| 2020             | 45         | 44   | 43   | 42   | 41   | 40   | 39   | 38   | 37   | 36   | 35   | 34   | 33   |
| 2021             | 46         | 45   | 44   | 43   | 42   | 41   | 40   | 39   | 38   | 37   | 36   | 35   | 34   |
| 2022             | 47         | 46   | 45   | 44   | 43   | 42   | 41   | 40   | 39   | 38   | 37   | 36   | 35   |
| 2023             | 48         | 47   | 46   | 45   | 44   | 43   | 42   | 41   | 40   | 39   | 38   | 37   | 36   |
| 2024             | 49         | 48   | 47   | 46   | 45   | 44   | 43   | 42   | 41   | 40   | 39   | 38   | 37   |

Age at SSN  
assignment

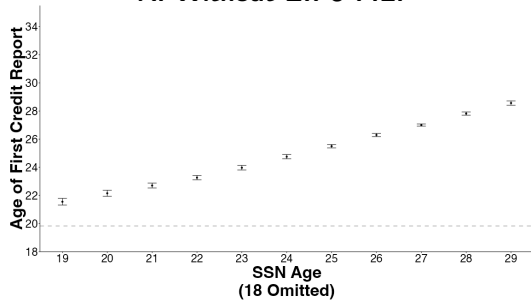
Years of  
data in our  
sample

# Summary Statistics

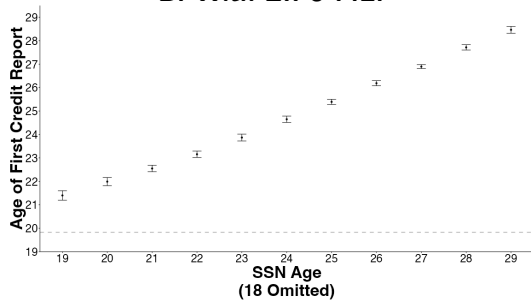
|                              | <u>Non-Immigrants</u><br><i>(SSN Age <math>\leq</math> 21)</i> |           | <u>Immigrants</u><br><i>(SSN Age 21 to 29)</i> |          |
|------------------------------|----------------------------------------------------------------|-----------|------------------------------------------------|----------|
|                              | Mean                                                           | <i>N</i>  | Mean                                           | <i>N</i> |
| <b>Credit Report</b>         |                                                                |           |                                                |          |
| Age at First Credit Report   | 19.86                                                          | 5,778,671 | 25.21                                          | 344,261  |
| <b>Credit Score</b>          |                                                                |           |                                                |          |
| Any Score by Age 30 (%)      | 97.57                                                          | 5,778,671 | 88.14                                          | 344,261  |
| Credit Score at Age 30       | 626.1                                                          | 5,461,785 | 660.7                                          | 293,349  |
| Any Score by Age 40 (%)      | 99.68                                                          | 4,487,804 | 99.14                                          | 312,391  |
| Credit Score at Age 40       | 659.5                                                          | 4,179,752 | 698.9                                          | 270,177  |
| <b>Auto Loan</b>             |                                                                |           |                                                |          |
| Any Auto Loan (%)            | 79.97                                                          | 5,778,671 | 66.07                                          | 344,261  |
| Age at First Auto Loan       | 27.62                                                          | 4,621,146 | 31.61                                          | 227,439  |
| <b>Mortgage</b>              |                                                                |           |                                                |          |
| Any Mortgage (%)             | 50.06                                                          | 5,778,671 | 46.41                                          | 344,261  |
| Age at First Mortgage        | 30.59                                                          | 2,892,536 | 33.79                                          | 159,758  |
| <b>Credit Card</b>           |                                                                |           |                                                |          |
| Any Credit Card (%)          | 94.51                                                          | 5,778,671 | 94.00                                          | 344,261  |
| Age at First Credit Card     | 24.42                                                          | 5,461,356 | 27.97                                          | 323,607  |
| Credit Card Limits at Age 30 | \$11,733                                                       | 5,778,671 | \$11,193                                       | 344,261  |
| Credit Card Limits at Age 40 | \$22,477                                                       | 4,487,804 | \$28,158                                       | 312,391  |

# Age At First Credit Report

**A. Without ZIP5 F.E.**

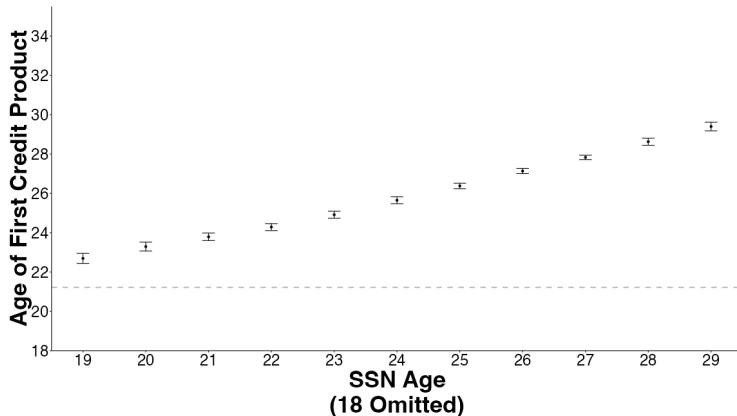


**B. With ZIP5 F.E.**



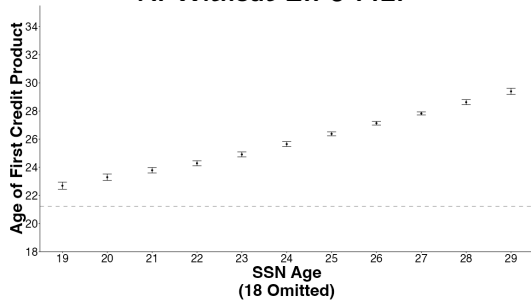
## SSN Assigned At Older Age $\rightarrow$ Later Entry Into U.S. Credit System

$$Y_i = \phi_{SSN\ Age(i)} + \mu_{b(i)} + \epsilon_i \quad (1)$$

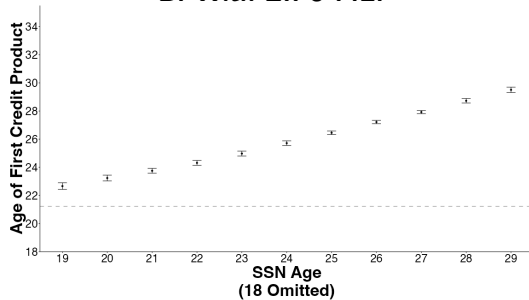


# Age At First Credit Product

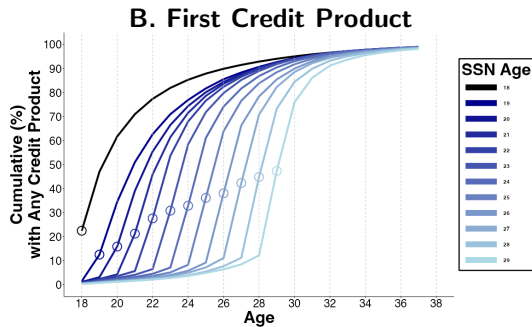
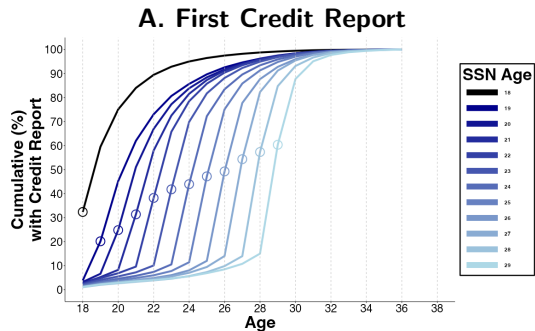
**A. Without ZIP5 F.E.**



**B. With ZIP5 F.E.**



# Age At First Credit Report and First Credit Product By SSN Age Cohort



## Credit Entry 🚀

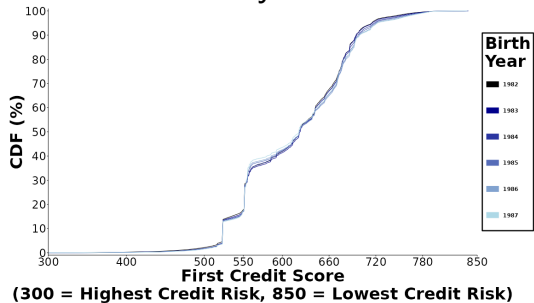
| Dep Var: Age at First... | Credit Report       |                     | Credit Product      |                     |
|--------------------------|---------------------|---------------------|---------------------|---------------------|
|                          | (1)                 | (2)                 | (3)                 | (4)                 |
| Immigrant                | 4.979***<br>(0.143) | 1.808***<br>(0.088) | 4.686***<br>(0.149) | 1.618***<br>(0.106) |
| SSN Age                  |                     | 0.744***<br>(0.014) |                     | 0.720***<br>(0.016) |
| F.E. Birth Year          | X                   | X                   | X                   | X                   |
| F.E. First Zip5          | X                   | X                   | X                   | X                   |
| $R^2$                    | 0.264               | 0.289               | 0.120               | 0.129               |
| $N$                      | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           |
| Mean, SSN Age j21        | 19.864              | 19.864              | 21.251              | 21.251              |

# Credit Entry With Additional F.E.

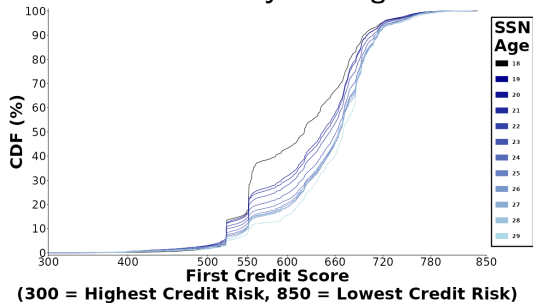
| Dep Var: Age at First... | <u>Credit Report</u> |                     | <u>Credit Product</u> |                     | <u>Auto Loan</u>    |                     | <u>Mortgage</u>     |                     | <u>Credit Card</u>  |                     |
|--------------------------|----------------------|---------------------|-----------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                          | (1)                  | (2)                 | (3)                   | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 | (9)                 | (10)                |
| Immigrant                | 4.782***<br>(0.117)  | 1.691***<br>(0.073) | 4.403***<br>(0.116)   | 1.477***<br>(0.088) | 3.719***<br>(0.206) | 1.424***<br>(0.136) | 1.642***<br>(0.101) | -0.082<br>(0.052)   | 3.929***<br>(0.146) | 1.195***<br>(0.125) |
| SSN Age                  |                      | 0.727***<br>(0.014) |                       | 0.689***<br>(0.016) |                     | 0.540***<br>(0.023) |                     | 0.406***<br>(0.017) |                     | 0.644***<br>(0.019) |
| F.E. Birth Year          | X                    | X                   | X                     | X                   | X                   | X                   | X                   | X                   | X                   | X                   |
| F.E. First Zip5          | X                    | X                   | X                     | X                   | X                   | X                   | X                   | X                   | X                   | X                   |
| F.E. Last Zip5           | X                    | X                   | X                     | X                   | X                   | X                   | X                   | X                   | X                   | X                   |
| F.E. Longest Zip5        | X                    | X                   | X                     | X                   | X                   | X                   | X                   | X                   | X                   | X                   |
| F.E. Number Zip5         | X                    | X                   | X                     | X                   | X                   | X                   | X                   | X                   | X                   | X                   |
| $R^2$                    | 0.299                | 0.323               | 0.184                 | 0.192               | 0.168               | 0.169               | 0.153               | 0.154               | 0.155               | 0.159               |
| $N$                      | 6,122,932            | 6,122,932           | 6,122,932             | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           |
| Mean, SSN Age j21        | 19.864               | 19.864              | 21.251                | 21.251              | 29.151              | 29.151              | 36.356              | 36.356              | 22.677              | 22.677              |

# CDF Of First Credit Score 🚀

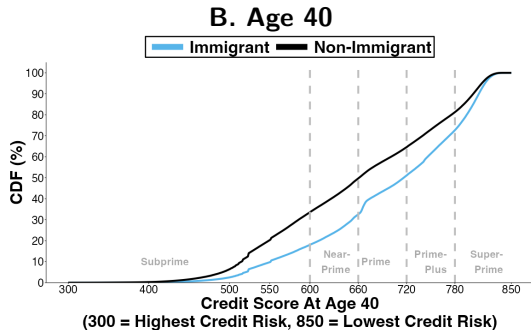
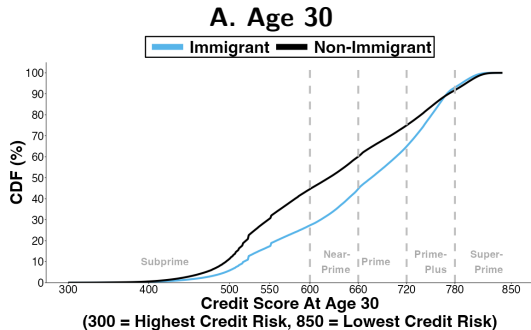
## A. CDF By Birth Year



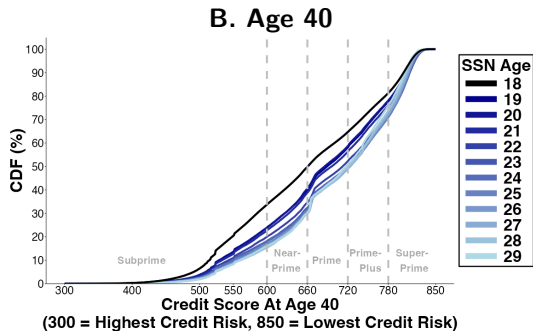
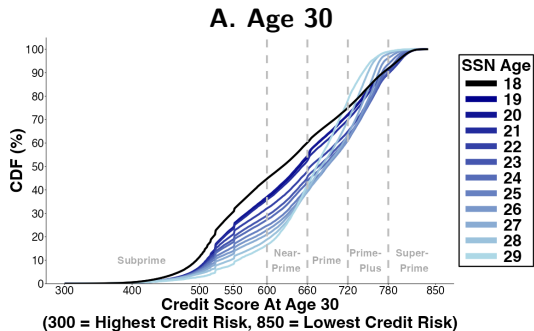
## B. CDF By SSN Age



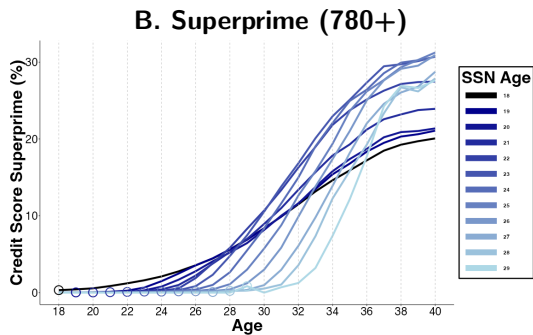
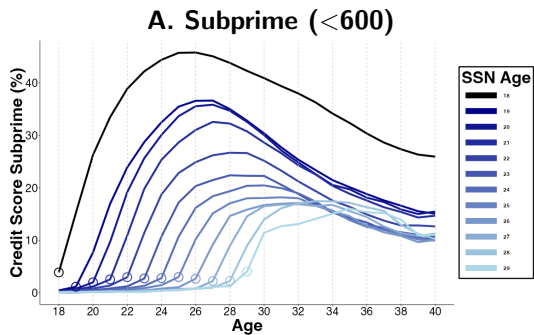
# CDF Of Credit Score At Ages 30 & 40, By Immigration Status 🗺️



# CDF Of Credit Score At Ages 30 & 40, By SSN Age 🗺



# Lifecycle Of The Distribution Of Credit Scores By SSN Age Cohorts 🚀



# Immigrant Credit Scores Higher In Spite Of Shorter Credit Histories

|                   | Credit Score at  |                  | Any Prime or Higher<br>(660+) Credit Score at |                 |
|-------------------|------------------|------------------|-----------------------------------------------|-----------------|
|                   | Age 30           | Age 40           | Age 30                                        | Age 40          |
|                   | (1)              | (2)              | (3)                                           | (4)             |
| Immigrant         | 15.7***<br>(1.2) | 21.6***<br>(0.6) | 6.6***<br>(0.6)                               | 2.0***<br>(0.2) |
| SSN Age           | 1.6***<br>(0.3)  | 1.5***<br>(0.2)  | -0.1<br>(0.2)                                 | 1.6***<br>(0.1) |
| F.E. Birth Year   | X                | X                | X                                             | X               |
| F.E. First Zip5   | X                | X                | X                                             | X               |
| $R^2$             | 0.202            | 0.200            | 0.156                                         | 0.152           |
| $N$               | 5,755,134        | 4,449,929        | 6,122,932                                     | 4,800,195       |
| Mean, SSN Age <21 | 626.1            | 659.5            | 37.71                                         | 46.84           |

# Credit Scores At Ages 30 & 40, 🇵🇸

| Dep Var: Credit Score at ... | Age 30           |                  | Age 40           |                  |
|------------------------------|------------------|------------------|------------------|------------------|
|                              | (1)              | (2)              | (3)              | (4)              |
| Immigrant                    | 27.1***<br>(1.1) | 18.8***<br>(1.6) | 32.7***<br>(1.0) | 25.8***<br>(0.7) |
| SSN Age                      |                  | 2.0***<br>(0.4)  |                  | 1.5***<br>(0.2)  |
| F.E. Birth Year              | X                | X                | X                | X                |
| F.E. First Zip5              | X                | X                | X                | X                |
| $R^2$                        | 0.135            | 0.135            | 0.132            | 0.132            |
| $N$                          | 5,755,134        | 5,755,134        | 4,449,929        | 4,449,929        |
| Mean, SSN Age j21            | 626.1            | 626.1            | 659.5            | 659.5            |

# Credit Scores At Ages 30 & 40, With Additional F.E. 📌

|                   | Credit Score At... |                    |                    |                    | Probability of Prime or Higher Score At... |                   |                   |                   |
|-------------------|--------------------|--------------------|--------------------|--------------------|--------------------------------------------|-------------------|-------------------|-------------------|
|                   | Age 30             |                    | Age 40             |                    | Age 30                                     |                   | Age 40            |                   |
|                   | (1)                | (2)                | (3)                | (4)                | (5)                                        | (6)               | (7)               | (8)               |
| Immigrant         | 22.27***<br>(1.03) | 15.69***<br>(1.24) | 28.14***<br>(0.94) | 21.56***<br>(0.64) | 5.99***<br>(0.92)                          | 6.60***<br>(0.57) | 8.97***<br>(0.27) | 1.99***<br>(0.21) |
| SSN Age           |                    | 1.61***<br>(0.31)  |                    | 1.46***<br>(0.16)  |                                            | -0.14<br>(0.19)   |                   | 1.58***<br>(0.09) |
| F.E. Birth Year   | X                  | X                  | X                  | X                  | X                                          | X                 | X                 | X                 |
| F.E. First Zip5   | X                  | X                  | X                  | X                  | X                                          | X                 | X                 | X                 |
| F.E. Last Zip5    | X                  | X                  | X                  | X                  | X                                          | X                 | X                 | X                 |
| F.E. Longest Zip5 | X                  | X                  | X                  | X                  | X                                          | X                 | X                 | X                 |
| F.E. Number Zip5  | X                  | X                  | X                  | X                  | X                                          | X                 | X                 | X                 |
| $R^2$             | 0.202              | 0.202              | 0.200              | 0.200              | 0.156                                      | 0.156             | 0.151             | 0.152             |
| $N$               | 5,755,134          | 5,755,134          | 4,449,929          | 4,449,929          | 6,122,932                                  | 6,122,932         | 4,800,195         | 4,800,195         |
| Mean, SSN Age j21 | 626.10             | 626.10             | 659.52             | 659.52             | 37.71                                      | 37.71             | 46.84             | 46.84             |

# Credit Score Prime or Higher (660+) At Ages 30 & 40, 🍷

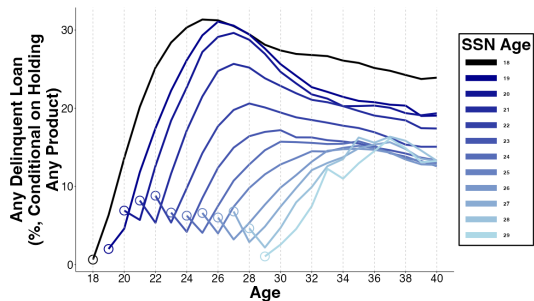
| Dep Var: Prime or Higher at ... | <u>Age 30</u>     |                   | <u>Age 40</u>     |                   |
|---------------------------------|-------------------|-------------------|-------------------|-------------------|
|                                 | (1)               | (2)               | (3)               | (4)               |
| Immigrant                       | 7.30***<br>(1.02) | 7.63***<br>(0.74) | 9.00***<br>(0.20) | 2.01***<br>(0.14) |
| SSN Age                         |                   | -0.08<br>(0.20)   |                   | 1.58***<br>(0.09) |
| F.E. Birth Year                 | X                 | X                 | X                 | X                 |
| F.E. First Zip5                 | X                 | X                 | X                 | X                 |
| $R^2$                           | 0.104             | 0.104             | 0.092             | 0.092             |
| $N$                             | 6,122,932         | 6,122,932         | 4,800,195         | 4,800,195         |
| Mean, SSN Age j21               | 37.71             | 37.71             | 46.84             | 46.84             |

**Credit Score Prime or Higher (660+) At Ages 30 & 40, With Additional F.E.**



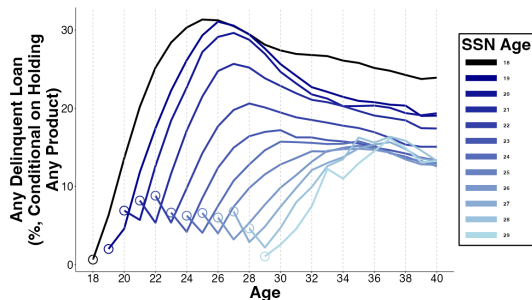
# Immigrants (vs. Non-Immigrants) Have More Creditworthy Behaviors: Lower Delinquency Rates & Lower Credit Card Utilization

## A. Any Delinquency

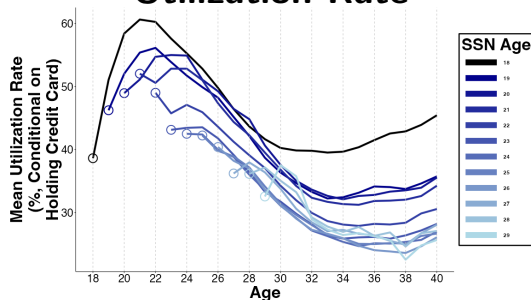


# Immigrants (vs. Non-Immigrants) Have More Creditworthy Behaviors: Lower Delinquency Rates & Lower Credit Card Utilization

## A. Any Delinquency



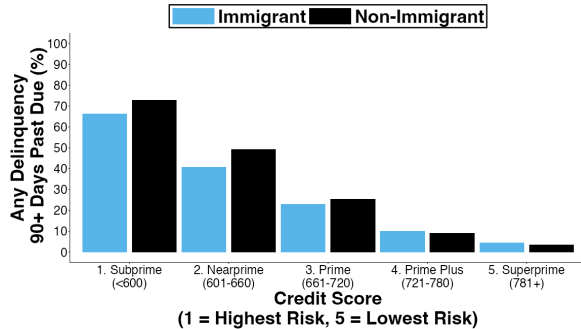
## B. Credit Card Utilization Rate



↓ 5% delinquency rates at age 37 conditional on credit score at age 30

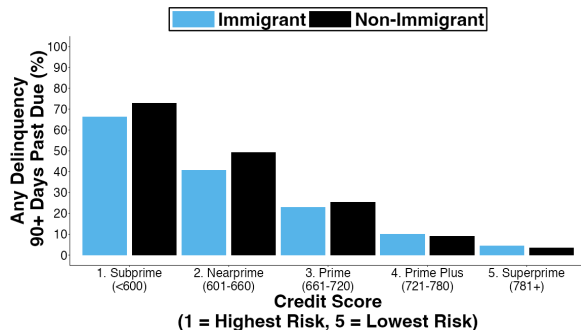
# Immigrants (vs. Non-Immigrants) Are More Creditworthy:

## 5% Lower Delinquency Rates At Age 37 Conditional On Age 30 Credit Score



# Immigrants (vs. Non-Immigrants) Are More Creditworthy:

## 5% Lower Delinquency Rates At Age 37 Conditional On Age 30 Credit Score



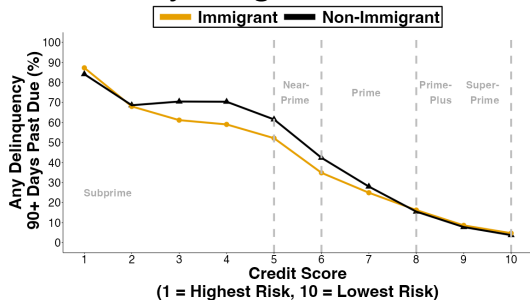
|                          |                        |
|--------------------------|------------------------|
| Immigrant                | -0.0140***<br>(0.0041) |
| SSN Age                  | -0.0004<br>(0.0005)    |
| F.E. Birth Year          | X                      |
| F.E. Age 30 Zip5         | X                      |
| F.E. Age 30 Credit Score | X                      |
| $R^2$                    | 0.185                  |
| $N$                      | 5,755,134              |
| Mean, SSN Age <21        | 0.2722                 |

# Calibration Bias: Delinquency At 37 Conditional On Age 30 Credit Score

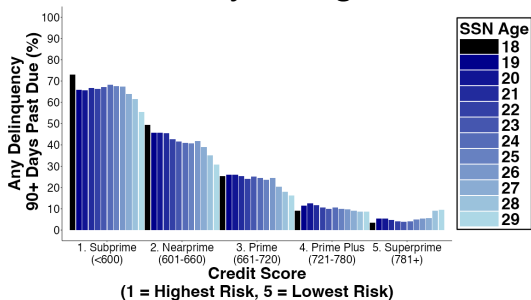
| Dep Var: Any<br>delinquency by age 37 | <u>All</u>         |                    | <u>Subprime</u>    |                    | <u>Nearprime</u>   |                    | <u>Prime</u>       |                    | <u>Prime Plus</u> |                    | <u>Superprime</u> |                 |
|---------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-------------------|--------------------|-------------------|-----------------|
|                                       | (1)                | (2)                | (3)                | (4)                | (5)                | (6)                | (7)                | (8)                | (9)               | (10)               | (11)              | (12)            |
| Immigrant                             | -1.58***<br>(0.24) | -1.40***<br>(0.41) | -2.85***<br>(0.40) | -3.69***<br>(0.74) | -3.53***<br>(0.36) | -2.72***<br>(0.58) | -1.49***<br>(0.24) | 0.06<br>(0.49)     | 0.08<br>(0.10)    | 0.99***<br>(0.19)  | 0.28<br>(0.14)    | 0.37<br>(0.28)  |
| SSN Age                               |                    | -0.04<br>(0.05)    |                    | 0.23<br>(0.12)     |                    | -0.19<br>(0.09)    |                    | -0.34***<br>(0.07) |                   | -0.21***<br>(0.03) |                   | -0.03<br>(0.07) |
| F.E. Birth Year                       | X                  | X                  | X                  | X                  | X                  | X                  | X                  | X                  | X                 | X                  | X                 | X               |
| F.E. Age 30 Zip5                      | X                  | X                  | X                  | X                  | X                  | X                  | X                  | X                  | X                 | X                  | X                 | X               |
| F.E. Age 30 Credit Score              | X                  | X                  | X                  | X                  | X                  | X                  | X                  | X                  | X                 | X                  | X                 | X               |
| $R^2$                                 | 0.185              | 0.185              | 0.032              | 0.032              | 0.028              | 0.028              | 0.027              | 0.027              | 0.025             | 0.025              | 0.022             | 0.022           |
| $N$                                   | 5,755,134          | 5,755,134          | 2,515,933          | 2,515,933          | 898,118            | 898,118            | 871,578            | 871,578            | 999,130           | 999,130            | 999,130           | 470,375         |
| Mean, SSN Age j21                     | 27.22              | 27.22              | 45.53              | 45.53              | 26.20              | 26.20              | 13.14              | 13.14              | 4.51              | 4.51               | 1.63              | 1.63            |

# Any Delinquency By Age 37, Conditional On Age 30 Credit Score 📍

## A. By Immigration Status



## B. By SSN Age

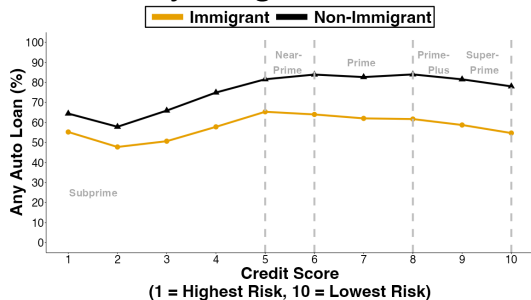


# Age 37 Credit Outcomes Conditional On Age 30 Credit Score 🍷

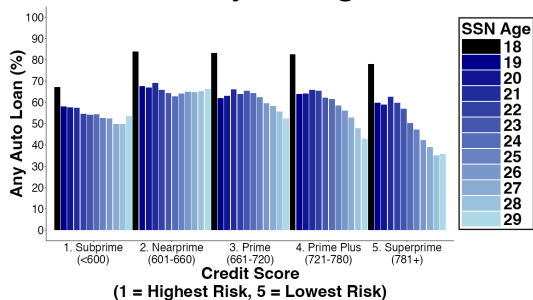
| Dep Var: Outcomes<br>at Age 37... | <u>Any<br/>Auto Loan</u> |                    | <u>Any<br/>Mortgage</u> |                    | <u>Any<br/>Credit Card</u> |                    | <u>Number of<br/>Credit Cards</u> |                   | <u>Credit<br/>Card Limits</u> |                     |
|-----------------------------------|--------------------------|--------------------|-------------------------|--------------------|----------------------------|--------------------|-----------------------------------|-------------------|-------------------------------|---------------------|
|                                   | (1)                      | (2)                | (3)                     | (4)                | (5)                        | (6)                | (7)                               | (8)               | (9)                           | (10)                |
| Immigrant                         | -14.68***<br>(0.49)      | -8.89***<br>(0.39) | -1.42***<br>(0.06)      | -1.61***<br>(0.13) | -11.93***<br>(0.49)        | -2.33***<br>(0.50) | 0.40***<br>(0.04)                 | 0.22***<br>(0.03) | -795.8<br>(565.0)             | 981.1<br>(561.8)    |
| SSN Age                           |                          | -1.42***<br>(0.06) |                         | 0.05<br>(0.03)     |                            | -2.35***<br>(0.12) |                                   | 0.04***<br>(0.01) |                               | -434.6**<br>(131.7) |
| F.E. Birth Year                   | X                        | X                  | X                       | X                  | X                          | X                  | X                                 | X                 | X                             | X                   |
| F.E. Age 30 Zip5                  | X                        | X                  | X                       | X                  | X                          | X                  | X                                 | X                 | X                             | X                   |
| F.E. Age 30 Credit Score          | X                        | X                  | X                       | X                  | X                          | X                  | X                                 | X                 | X                             | X                   |
| $R^2$                             | 0.107                    | 0.108              | 0.268                   | 0.268              | 0.110                      | 0.110              | 0.142                             | 0.142             | 0.306                         | 0.306               |
| $N$                               | 5,755,134                | 5,755,134          | 5,755,134               | 5,755,134          | 5,755,134                  | 5,755,134          | 5,755,134                         | 5,755,134         | 5,755,134                     | 5,755,134           |
| Mean, SSN Age j21                 | 75.34                    | 75.34              | 44.73                   | 44.73              | 92.21                      | 92.21              | 3.92                              | 3.92              | \$20,515.6                    | \$20,515.6          |

# Any Auto Loan By Age 37, Conditional On Age 30 Credit Score 🚀

## A. By Immigration Status

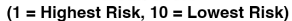


## B. By SSN Age

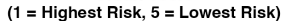




■ Immigrant ■ Non-Immigrant

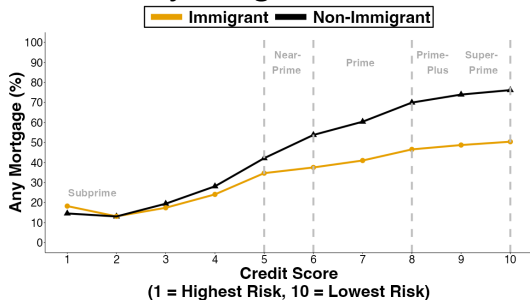


### B. By SSN Age

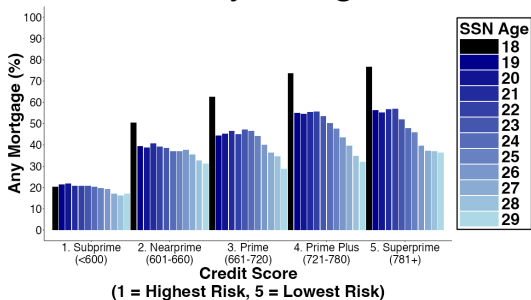


# Any Mortgage By Age 37, Conditional On Age 30 Credit Score 🚀

## A. By Immigration Status

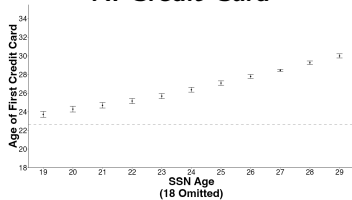


## B. By SSN Age

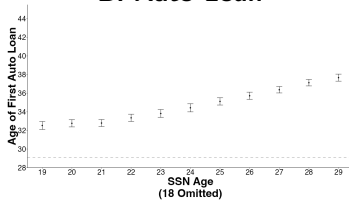


# Age At First Credit Product By Type 🚀

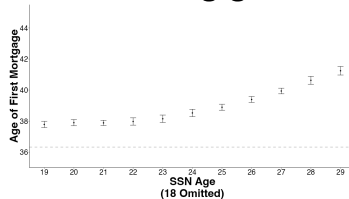
## A. Credit Card



## B. Auto Loan



## C. Mortgage



| Dep Var: Age at First... | <u>Auto Loan</u>    |                     | <u>Mortgage</u>     |                     | <u>Credit Card</u>  |                     |
|--------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                          | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
| Immigrant                | 4.604***<br>(0.274) | 1.967***<br>(0.217) | 2.058***<br>(0.151) | 0.219*<br>(0.083)   | 4.309***<br>(0.184) | 1.367***<br>(0.158) |
| SSN Age                  |                     | 0.619***<br>(0.024) |                     | 0.432***<br>(0.018) |                     | 0.690***<br>(0.020) |
| F.E. Birth Year          | X                   | X                   | X                   | X                   | X                   | X                   |
| F.E. First Zip5          | X                   | X                   | X                   | X                   | X                   | X                   |
| $R^2$                    | 0.080               | 0.082               | 0.080               | 0.081               | 0.080               | 0.085               |
| $N$                      | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           |
| Mean, SSN Age j21        | 29.151              | 29.151              | 36.356              | 36.356              | 22.677              | 22.677              |

# Credit Access With Additional F.E.

| Dep Var: Age at First... | Credit Report       |                     | Credit Product      |                     | Auto Loan           |                     | Mortgage            |                     | Credit Card         |                     |
|--------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                          | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 | (9)                 | (10)                |
| Immigrant                | 4.782***<br>(0.117) | 1.691***<br>(0.073) | 4.403***<br>(0.116) | 1.477***<br>(0.088) | 3.719***<br>(0.206) | 1.424***<br>(0.136) | 1.642***<br>(0.101) | -0.082<br>(0.052)   | 3.929***<br>(0.146) | 1.195***<br>(0.125) |
| SSN Age                  |                     | 0.727***<br>(0.014) |                     | 0.689***<br>(0.016) |                     | 0.540***<br>(0.023) |                     | 0.406***<br>(0.017) |                     | 0.644***<br>(0.019) |
| F.E. Birth Year          | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                   |
| F.E. First Zip5          | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                   |
| F.E. Last Zip5           | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                   |
| F.E. Longest Zip5        | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                   |
| F.E. Number Zip5         | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                   | X                   |
| $R^2$                    | 0.299               | 0.323               | 0.184               | 0.192               | 0.168               | 0.169               | 0.153               | 0.154               | 0.155               | 0.159               |
| $N$                      | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           | 6,122,932           |
| Mean, SSN Age j21        | 19.864              | 19.864              | 21.251              | 21.251              | 29.151              | 29.151              | 36.356              | 36.356              | 22.677              | 22.677              |

## Credit Access By Age 37

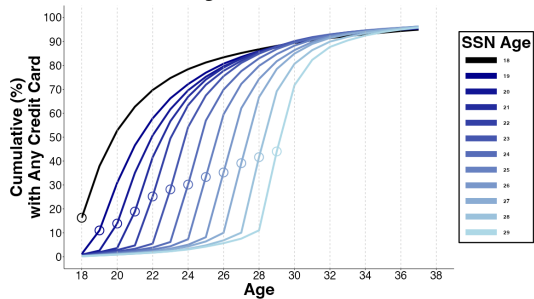
| Dep. Var: By Age 37, has ... | <u>Auto Loan</u>    |                    | <u>Mortgage</u>    |                    | <u>Credit Card</u> |                |
|------------------------------|---------------------|--------------------|--------------------|--------------------|--------------------|----------------|
|                              | (1)                 | (2)                | (3)                | (4)                | (5)                | (6)            |
| Immigrant                    | -12.99***<br>(0.80) | -6.87***<br>(0.46) | -7.18***<br>(0.74) | 0.39<br>(0.53)     | 0.31<br>(0.22)     | 0.15<br>(0.16) |
| SSN Age                      |                     | -1.44***<br>(0.10) |                    | -1.77***<br>(0.09) |                    | 0.04<br>(0.03) |
| F.E. Birth Year              | X                   | X                  | X                  | X                  | X                  | X              |
| F.E. First Zip5              | X                   | X                  | X                  | X                  | X                  | X              |
| $R^2$                        | 0.039               | 0.039              | 0.062              | 0.063              | 0.025              | 0.025          |
| $N$                          | 6,122,932           | 6,122,932          | 6,122,932          | 6,122,932          | 6,122,932          | 6,122,932      |
| Mean, SSN Age j21            | 77.22               | 77.22              | 45.46              | 45.46              | 94.93              | 94.93          |

# Credit Access By Age 37 With Additional F.E.

| Dep Var: By Age 37, has...   | <u>Auto Loan</u>   |                    | <u>Mortgage</u>    |                    | <u>Credit Card</u> |                   |
|------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-------------------|
|                              | (1)                | (2)                | (3)                | (4)                | (5)                | (6)               |
| Immigrant                    | -8.94***<br>(0.67) | -4.43***<br>(0.20) | -4.30***<br>(0.67) | 2.33***<br>(0.35)  | 1.14***<br>(0.22)  | 0.50***<br>(0.14) |
| SSN Age                      |                    | -1.06***<br>(0.11) |                    | -1.56***<br>(0.09) |                    | 0.15***<br>(0.04) |
| F.E. Birth Year              | X                  | X                  | X                  | X                  | X                  | X                 |
| F.E. First Zip5              | X                  | X                  | X                  | X                  | X                  | X                 |
| F.E. Last Zip5               | X                  | X                  | X                  | X                  | X                  | X                 |
| F.E. Longest Zip5            | X                  | X                  | X                  | X                  | X                  | X                 |
| F.E. Number Zip5             | X                  | X                  | X                  | X                  | X                  | X                 |
| $R^2$                        | 0.124              | 0.124              | 0.146              | 0.146              | 0.060              | 0.060             |
| $N$                          | 6,122,932          | 6,122,932          | 6,122,932          | 6,122,932          | 6,122,932          | 6,122,932         |
| Mean, SSN Age $\bar{x}_{21}$ | 77.22              | 77.22              | 45.46              | 45.46              | 94.93              | 94.93             |

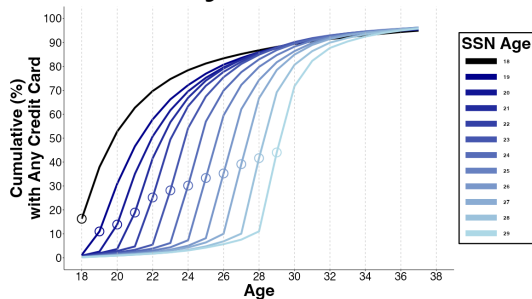
# Immigrants' Access to Credit Cards Converges With Non-Immigrants

## A. Any Credit Card

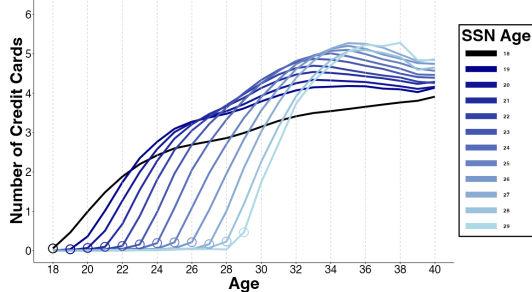


# Immigrants' Access to Credit Cards Converges With Non-Immigrants

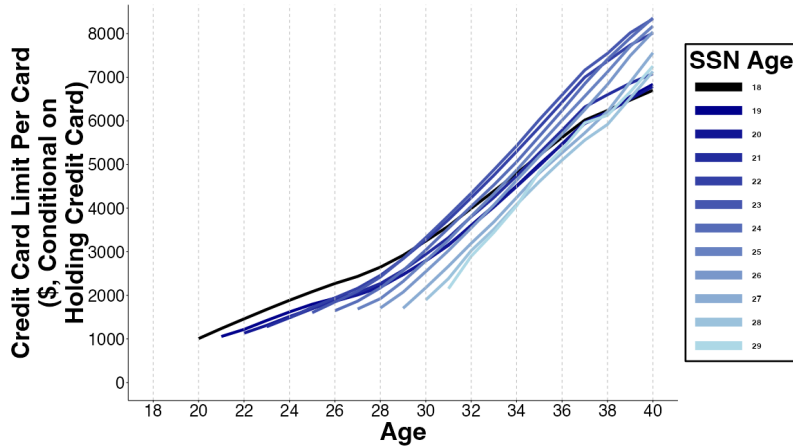
## A. Any Credit Card



## B. Number of Credit Cards

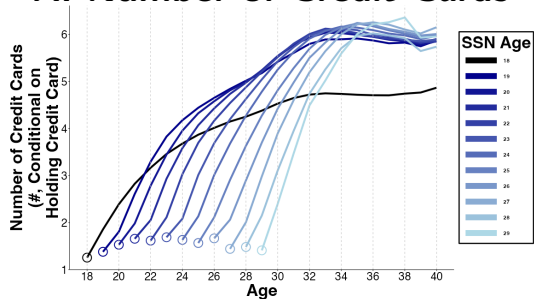


# Credit Card Limit Per Card 🚀



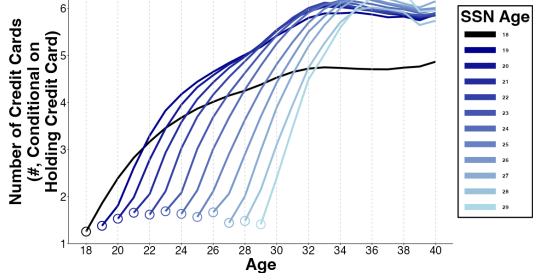
# Immigrants' Credit Card Access, Conditional On Holding Any Credit Card

## A. Number of Credit Cards

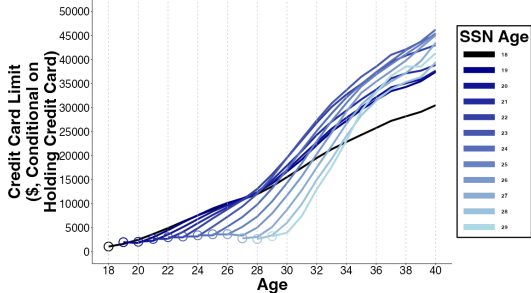


## Immigrants' Credit Card Access, Conditional On Holding Any Credit Card

### A. Number of Credit Cards



## B. Credit Card Limits



## Number of Credit Cards At Age 30 🚀

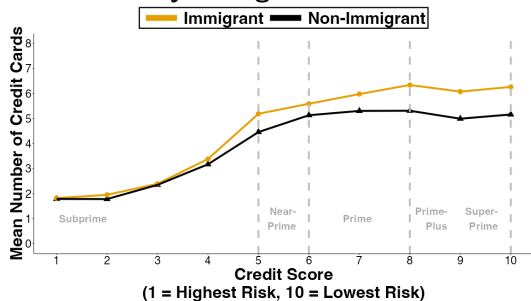
## Number of Credit Cards At Age 40 🚀

## Credit Card Limits At Age 30 🚀

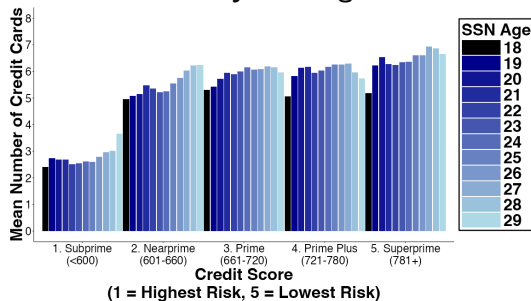
## Credit Card Limits At Age 40 🚀

# Number Of Credit Cards At Age 37, Conditional On Age 30 Credit Score 📍

## A. By Immigration Status

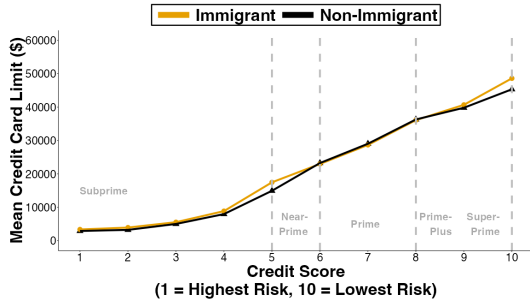


## B. By SSN Age

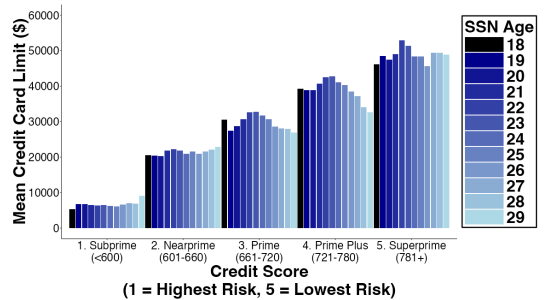


# Credit Card Limits At Age 37, Conditional On Age 30 Credit Score 🚀

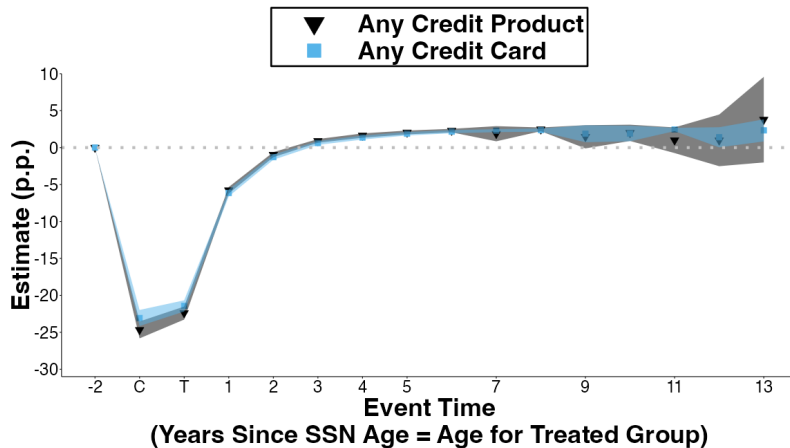
## A. By Immigration Status



## B. By SSN Age

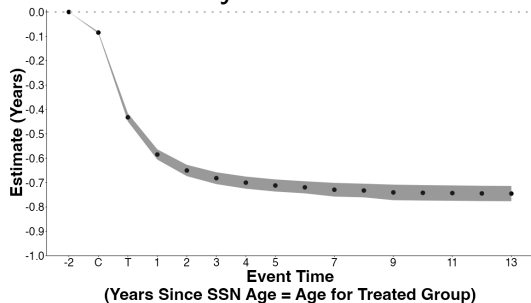


## Paired Cohorts: First Credit Product & First Credit Card 🚀

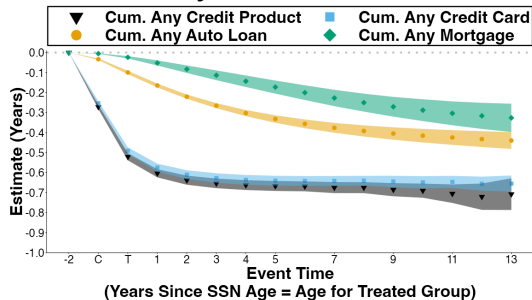


# Paired Cohorts: Cumulatives 🚀

## A. Any Credit Score



## B. By Credit Product

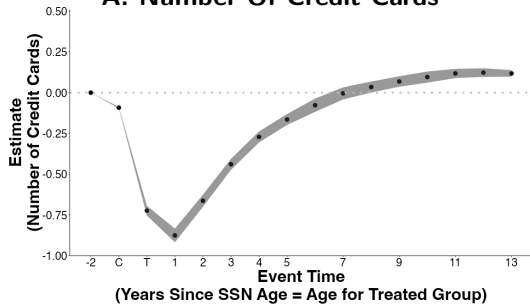


## Paired Cohorts: Cumulative Years of Delay 🚀

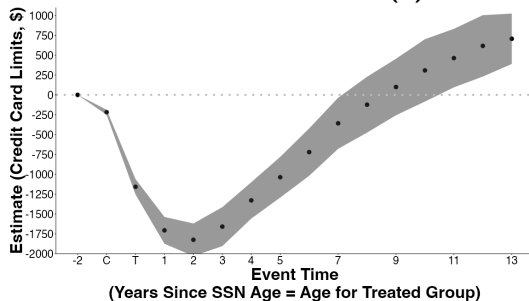
| Event Time              | <u>Any Credit Score</u><br>(1) | <u>Credit Product</u><br>(2) | <u>Auto Loan</u><br>(3) | <u>Mortgage</u><br>(4) | <u>Credit Card</u><br>(5) |
|-------------------------|--------------------------------|------------------------------|-------------------------|------------------------|---------------------------|
| C                       | -0.085***<br>(0.002)           | -0.272***<br>(0.005)         | -0.033***<br>(0.002)    | -0.005***<br>(0.001)   | -0.253***<br>(0.005)      |
| T                       | -0.432***<br>(0.009)           | -0.521***<br>(0.009)         | -0.099***<br>(0.004)    | -0.024***<br>(0.003)   | -0.491***<br>(0.008)      |
| T+5                     | -0.712***<br>(0.013)           | -0.667***<br>(0.013)         | -0.332***<br>(0.01)     | -0.173***<br>(0.017)   | -0.640***<br>(0.013)      |
| T+10                    | -0.742***<br>(0.016)           | -0.690***<br>(0.018)         | -0.415***<br>(0.017)    | -0.288***<br>(0.028)   | -0.648***<br>(0.016)      |
| <i>N</i> Paired Cohorts | 68                             | 68                           | 68                      | 68                     | 68                        |
| <i>N</i> Consumers      | 403,506                        | 403,506                      | 403,506                 | 403,506                | 403,506                   |
| <i>N</i>                | 6,456,096                      | 6,456,096                    | 6,456,096               | 6,456,096              | 6,456,096                 |

# Paired Cohorts: Credit Cards 🚀

## A. Number Of Credit Cards



## B. Credit Card Limits (\$)



## Potential Mechanism #2: Emigration

Proxy emigration by **attrition** from data

Immigrants exit data more than Non-Immigrants

- If arrive earlier, on average, exit data earlier, so can't explain paired cohort results

## Potential Mechanism #2: Emigration

Proxy emigration by **attrition** from data

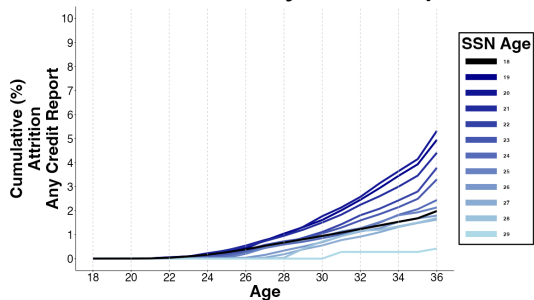
Immigrants exit data more than Non-Immigrants

- If arrive earlier, on average, exit data earlier, so can't explain paired cohort results

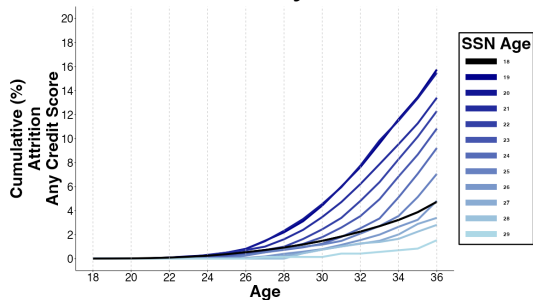
Restrict to consumers observed in 2024:

- **Mortgage** Immigrant vs. Non-Immigrant gap closes but SSN Age gap remains
- **Auto loan** Immigrant vs. Non-Immigrant gap & SSN Age gap remains
- **Credit card** dynamics by SSN Age remains

## A. Attrition In Any Credit Report

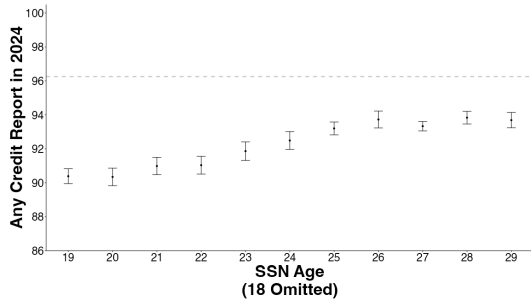


## B. Attrition In Any Credit Score

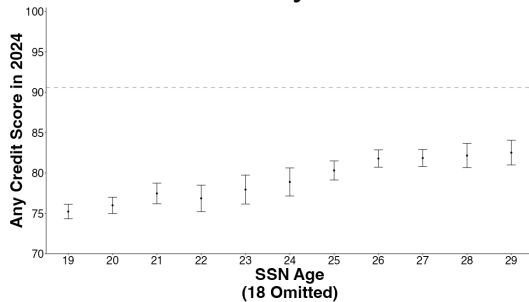


# Consumers Observed In 2024 🚀

## A. Attrition In Any Credit Report

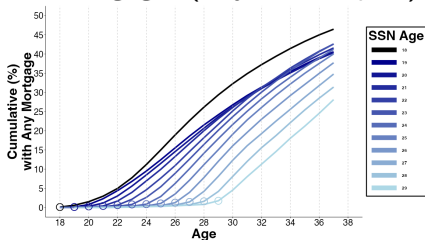


## B. Attrition In Any Credit Score

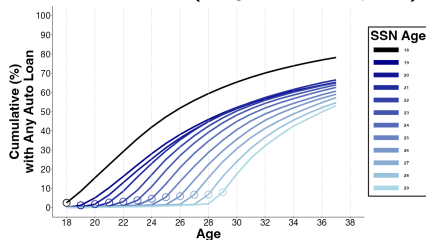


# Lifecycle Of Credit For Consumers Observed In 2024 📍

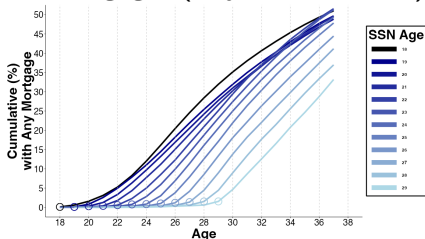
## A. Mortgages (Any 2024 Report)



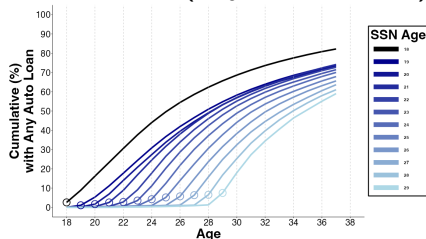
## B. Auto Loan (Any 2024 Report)



## C. Mortgages (Any 2024 Product)

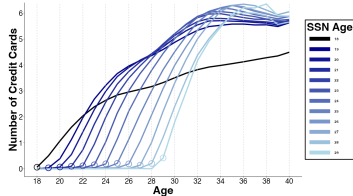


## D. Auto Loan (Any 2024 Product)

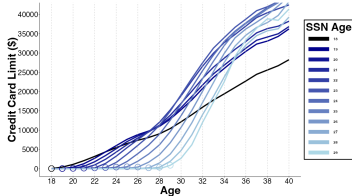


# Lifecycle Of Credit For Consumers Observed In 2024 (Any Tradeline) 🚀

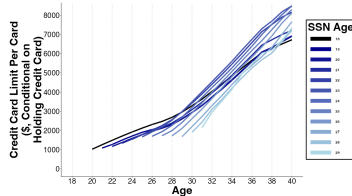
## A. Number Credit Cards



## B. Credit Card Limits



## C. Credit Limit Per Card



# Consumers Observed In Data In 2024 With Additional F.E. 📍

| Dep Var: In 2024, has... | Any Credit Report  |                    | Any Credit Score   |                     | Any Credit Tradeline |                     |
|--------------------------|--------------------|--------------------|--------------------|---------------------|----------------------|---------------------|
|                          | (1)                | (2)                | (3)                | (4)                 | (5)                  | (6)                 |
| Immigrant                | -2.64***<br>(0.13) | -4.85***<br>(0.22) | -7.52***<br>(0.29) | -12.16***<br>(0.48) | -7.50***<br>(0.32)   | -12.07***<br>(0.42) |
| SSN Age                  |                    | 0.52***<br>(0.04)  |                    | 1.09***<br>(0.07)   |                      | 1.08***<br>(0.06)   |
| F.E. Birth Year          | X                  | X                  | X                  | X                   | X                    | X                   |
| F.E. First Zip5          | X                  | X                  | X                  | X                   | X                    | X                   |
| F.E. Last Zip5           | X                  | X                  | X                  | X                   | X                    | X                   |
| F.E. Longest Zip5        | X                  | X                  | X                  | X                   | X                    | X                   |
| F.E. Number Zip5         | X                  | X                  | X                  | X                   | X                    | X                   |
| $R^2$                    | 0.034              | 0.034              | 0.110              | 0.111               | 0.138                | 0.138               |
| $N$                      | 6,122,932          | 6,122,932          | 6,122,932          | 6,122,932           | 6,122,932            | 6,122,932           |
| Mean, SSN Age j21        | 96.15              | 96.15              | 90.31              | 90.31               | 85.03                | 85.03               |

# Credit Access By Age 37 With Additional F.E. Conditional On Any Credit Score Observed in 2024 📍

| Dep Var: By Age 37, has...    | <u>Auto Loan</u>   |                    | <u>Mortgage</u>    |                    | <u>Credit Card</u> |                   |
|-------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-------------------|
|                               | (1)                | (2)                | (3)                | (4)                | (5)                | (6)               |
| Immigrant                     | -8.30***<br>(0.81) | -2.01***<br>(0.22) | -2.89***<br>(0.71) | 6.57***<br>(0.41)  | 1.03***<br>(0.20)  | 0.67***<br>(0.13) |
| SSN Age                       |                    | -1.46***<br>(0.12) |                    | -2.20***<br>(0.11) |                    | 0.08<br>(0.04)    |
| F.E. Birth Year               | X                  | X                  | X                  | X                  | X                  | X                 |
| F.E. First Zip5               | X                  | X                  | X                  | X                  | X                  | X                 |
| F.E. Last Zip5                | X                  | X                  | X                  | X                  | X                  | X                 |
| F.E. Longest Zip5             | X                  | X                  | X                  | X                  | X                  | X                 |
| F.E. Number Zip5              | X                  | X                  | X                  | X                  | X                  | X                 |
| $R^2$                         | 0.099              | 0.099              | 0.134              | 0.134              | 0.054              | 0.054             |
| $N$                           | 5,489,726          | 5,489,726          | 5,489,726          | 5,489,726          | 5,489,726          | 5,489,726         |
| Mean, SSN Age $\downarrow$ 21 | 80.37              | 80.37              | 48.76              | 48.76              | 96.05              | 96.05             |

# Use Survey of Consumer Finances (SCF) 2022 To Evaluate Mechanisms

SCF 2022 includes if Immigrant & Age of Immigration

Adapt earlier cross-sectional regression to this data:

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \beta_2 \cdot Immigration\ Age_i + \mu_{b(i)} + \eta_{d(i)} + \epsilon_i$$

$\eta_{d(i)}$  are F.E. Demographic Controls: male, race, hispanic, spouse, male spouse, marital status, household size, number of children, education levels, & quintiles of household wage income

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$\eta_{d(i)}$  are F.E. Demographic Controls: male, race, hispanic, spouse, male spouse, marital status, household size, number of children, education levels, & quintiles of household wage income

## SCF Debt Attitude Questions:

*“People have many different reasons for borrowing money which they pay back over a period of time. For each of the reasons I read, please tell me whether you feel it is all right for someone like yourself to borrow money...to finance the purchase of a car.”*

# Immigrants More Likely To Buy Cars With Cash (SCF, 2022)

## 12.5 p.p. [95% C.I. -2.2, 27.3], 17% Vs. Non-Immigrants

| Dep Var:                  | <u>Has Car</u> |        | <u>Any Auto Debt</u> |        | <u>Homeowner</u> |          | <u>Any Mortgage</u> |        | <u>Credit Card Limits</u> |            |
|---------------------------|----------------|--------|----------------------|--------|------------------|----------|---------------------|--------|---------------------------|------------|
|                           | (1)            | (2)    | (3)                  | (4)    | (5)              | (6)      | (7)                 | (8)    | (9)                       | (10)       |
| Immigrant                 | -7.13*         | -6.14  | -4.27                | -11.14 | -13.55***        | 0.35     | -13.09***           | -5.27  | -3,078.8                  | 6,373.8    |
|                           | (3.15)         | (4.96) | (5.08)               | (8.27) | (4.64)           | (6.85)   | (4.64)              | (7.22) | (3,460.2)                 | (6,316.0)  |
| Immigration Age           |                | -0.13  |                      | 0.88   |                  | -1.78*** |                     | -1.00  |                           | -1,213.9*  |
|                           |                | (0.47) |                      | (0.76) |                  | (0.59)   |                     | (0.64) |                           | (582.4)    |
| Demographic Controls      | X              | X      | X                    | X      | X                | X        | X                   | X      | X                         | X          |
| $R^2$                     | 0.110          | 0.110  | 0.063                | 0.063  | 0.258            | 0.261    | 0.258               | 0.259  | 0.254                     | 0.256      |
| $N$                       | 1,813          | 1,813  | 1,813                | 1,813  | 1,813            | 1,813    | 1,813               | 1,813  | 1,813                     | 1,813      |
| Mean, Immigration Age j21 | 89.23          | 89.23  | 47.14                | 47.14  | 53.98            | 53.98    | 45.25               | 45.25  | \$20,275.9                | \$20,275.9 |

\*\*\* 1%, \*\*, 5%, \* 10% statistical significance

# Immigrants Have ↓ Demand for Debt (↑ Auto, ↑ Credit Cards, ↑ Mortgage. All Attenuated By Immigration Age) To Non-Immigrants (SCF, 2022) ...But Low Statistical Power To Detect Out Differences

| Dep Var: Apply For...           | <u>Any Credit</u> |                 | <u>Auto</u>    |                 | <u>Mortgage</u> |                 | <u>Credit Card</u> |                 | <u>Other Credit</u> |                              |
|---------------------------------|-------------------|-----------------|----------------|-----------------|-----------------|-----------------|--------------------|-----------------|---------------------|------------------------------|
|                                 | (1)               | (2)             | (3)            | (4)             | (5)             | (6)             | (7)                | (8)             | (9)                 | (10)                         |
| Immigrant                       | 2.32<br>(4.56)    | -1.08<br>(7.17) | 2.03<br>(4.37) | -6.55<br>(7.53) | 2.11<br>(3.8)   | 5.65<br>(6.85)  | 1.63<br>(3.22)     | 2.02<br>(5.32)  | 1.84<br>(3.62)      | -7.66 <sup>+</sup><br>(4.61) |
| Immigration Age                 |                   | 0.44<br>(0.62)  |                | 1.10<br>(0.72)  |                 | -0.45<br>(0.65) |                    | -0.05<br>(0.48) |                     | 1.22*<br>(0.55)              |
| Demographic Controls            | X                 | X               | X              | X               | X               | X               | X                  | X               | X                   | X                            |
| $R^2$                           | 0.051             | 0.051           | 0.051          | 0.053           | 0.045           | 0.045           | 0.045              | 0.045           | 0.026               | 0.029                        |
| $N$                             | 1,813             | 1,813           | 1,813          | 1,813           | 1,813           | 1,813           | 1,813              | 1,813           | 1,813               | 1,813                        |
| Mean, Immigration Age $\geq 21$ | 64.81             | 64.81           | 19.95          | 19.95           | 11.28           | 11.28           | 9.28               | 9.28            | 12.08               | 12.08                        |

\*\*\* 1%, \*\* , 5%, \* 10% statistical significance

# Immigrants Have ↓ Demand for Debt (↑ Auto, ↑ Credit Cards, ↑ Mortgage. All Attenuated By Immigration Age) To Non-Immigrants (SCF, 2022) ...But Low Statistical Power To Detect Out Differences

| Dep Var: Apply For...           | <u>Any Credit</u> |                 | <u>Auto</u>    |                 | <u>Mortgage</u> |                 | <u>Credit Card</u> |                 | <u>Other Credit</u> |                              |
|---------------------------------|-------------------|-----------------|----------------|-----------------|-----------------|-----------------|--------------------|-----------------|---------------------|------------------------------|
|                                 | (1)               | (2)             | (3)            | (4)             | (5)             | (6)             | (7)                | (8)             | (9)                 | (10)                         |
| Immigrant                       | 2.32<br>(4.56)    | -1.08<br>(7.17) | 2.03<br>(4.37) | -6.55<br>(7.53) | 2.11<br>(3.8)   | 5.65<br>(6.85)  | 1.63<br>(3.22)     | 2.02<br>(5.32)  | 1.84<br>(3.62)      | -7.66 <sup>+</sup><br>(4.61) |
| Immigration Age                 |                   | 0.44<br>(0.62)  |                | 1.10<br>(0.72)  |                 | -0.45<br>(0.65) |                    | -0.05<br>(0.48) |                     | 1.22*<br>(0.55)              |
| Demographic Controls            | X                 | X               | X              | X               | X               | X               | X                  | X               | X                   | X                            |
| $R^2$                           | 0.051             | 0.051           | 0.051          | 0.053           | 0.045           | 0.045           | 0.045              | 0.045           | 0.026               | 0.029                        |
| $N$                             | 1,813             | 1,813           | 1,813          | 1,813           | 1,813           | 1,813           | 1,813              | 1,813           | 1,813               | 1,813                        |
| Mean, Immigration Age $\geq 21$ | 64.81             | 64.81           | 19.95          | 19.95           | 11.28           | 11.28           | 9.28               | 9.28            | 12.08               | 12.08                        |

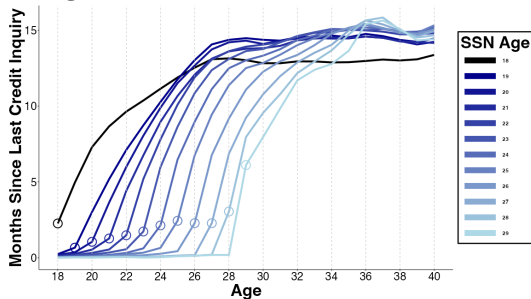
\*\*\* 1%, \*\* , 5%, \* 10% statistical significance

**Credit inquiries data in credit panel shows**  
**↑ immigrant credit demand in 20s, ↓ demand in 30s**

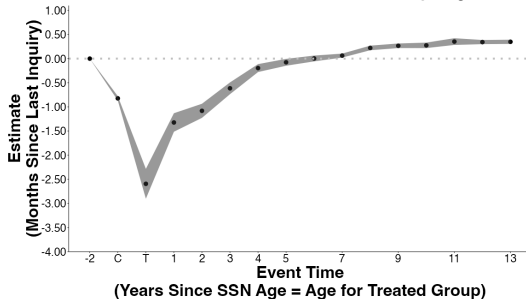
# Does Lower Credit Demand Explain Lower Access?

Immigrants Have  $\uparrow$  Credit Demand In 20s,  $\downarrow$  Demand In 30s 🚀

**A. Months Since Last Credit Inquiry**  
Higher Value = Lower Credit Demand



**B. Paired Cohorts**  
Months Since Last Credit Inquiry



**Caveat:** Lower credit demand in 30s may be function of earlier rejections  $\rightarrow$  deterrence

# Inconclusive Evidence On Other Demand and Supply Proxies (SCF, 2022)

## ...Given Low Statistical Power To Detect Out Differences

| (Conditional On...<br>Dep Var:  | Why Did Not Apply For Credit? |                 |                |                 | Outcome After Applying For Credit? |                 |                 |                   |                      |                    |
|---------------------------------|-------------------------------|-----------------|----------------|-----------------|------------------------------------|-----------------|-----------------|-------------------|----------------------|--------------------|
|                                 | ...Not Applying)              |                 |                |                 | ...Applying)                       |                 | ...Reject)      |                   | ...Reject & Reapply) |                    |
|                                 | No Demand                     |                 | Rate Too High  |                 | Reject/Less                        |                 | Reapply         |                   | Reapply Reject       |                    |
|                                 | (1)                           | (2)             | (3)            | (4)             | (5)                                | (6)             | (7)             | (8)               | (9)                  | (10)               |
| Immigrant                       | 0.41<br>(7.98)                | 6.84<br>(12.64) | 8.01<br>(5.85) | -0.73<br>(8.25) | -4.95<br>(4.74)                    | -2.28<br>(7.43) | 3.43<br>(14.11) | -10.42<br>(22.95) | 46.10***<br>(14.27)  | 49.69**<br>(18.81) |
| Immigration Age                 |                               | -0.88<br>(1.03) |                | 1.20<br>(0.87)  |                                    | -0.33<br>(0.61) |                 | 1.73<br>(1.89)    |                      | -0.40<br>(1.68)    |
| Demographic Controls            | X                             | X               | X              | X               | X                                  | X               | X               | X                 | X                    | X                  |
| $R^2$                           | 0.155                         | 0.155           | 0.090          | 0.092           | 0.054                              | 0.054           | 0.063           | 0.062             | 0.312                | 0.307              |
| $N$                             | 666                           | 666             | 666            | 666             | 1,147                              | 1,147           | 292             | 292               | 178                  | 178                |
| Mean, Immigration Age $\geq 21$ | 79.52                         | 79.52           | 7.53           | 7.53            | 27.07                              | 27.07           | 60.30           | 60.30             | 60.22                | 60.22              |

\*\*\* 1%, \*\*, 5%, \* 10% statistical significance